RING MAPS OUT OF A FINITE-DIMENSIONAL REGULAR RING ARE PERFECT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- A regular ring of finite dimension has finite global dimension. invoked at proof lines 21841-21843. Provides the finite global dimension of A.
- Over a ring of finite global dimension every module has finite tor dimension. invoked at proof lines 21844-21846. Gives the finite tor dimension of B.
- A finite type ring map out of a Noetherian ring is pseudo-coherent. invoked at proof lines 21847-21848. Gives the pseudo-coherence half of perfectness.

Source reference. Original lemma 00OE: sources/stacks-chow/tags.txt, source lines 892–892. Cited at proof lines 21841-21843 for the finite global dimension of A; the conclusion is used inside the counted span.

Source reference. Original lemma 066P: sources/stacks-more-algebra/pinned-original-source.tex, source lines 17287–17287. Cited at proof lines 21844-21845 for the finite tor dimension of B; the conclusion is used inside the counted span.

Source reference. Original lemma 067I: sources/stacks-more-algebra/pinned-original-source.tex, source lines 21805–21805. Cited at proof lines 21847-21848 for the pseudo-coherence of A -> B; the conclusion is used inside the counted span.

Lemma 0.2. *Let $A \to B$ be a finite type ring map with A a regular ring of finite dimension. Then $A \to B$ is perfect.*

Proof. By Algebra, Lemma 00OE the assumption on A means that A has finite global dimension. Hence every module has finite tor dimension, see Lemma 066P, in particular B does. By Lemma 067I the map is pseudo-coherent. □

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